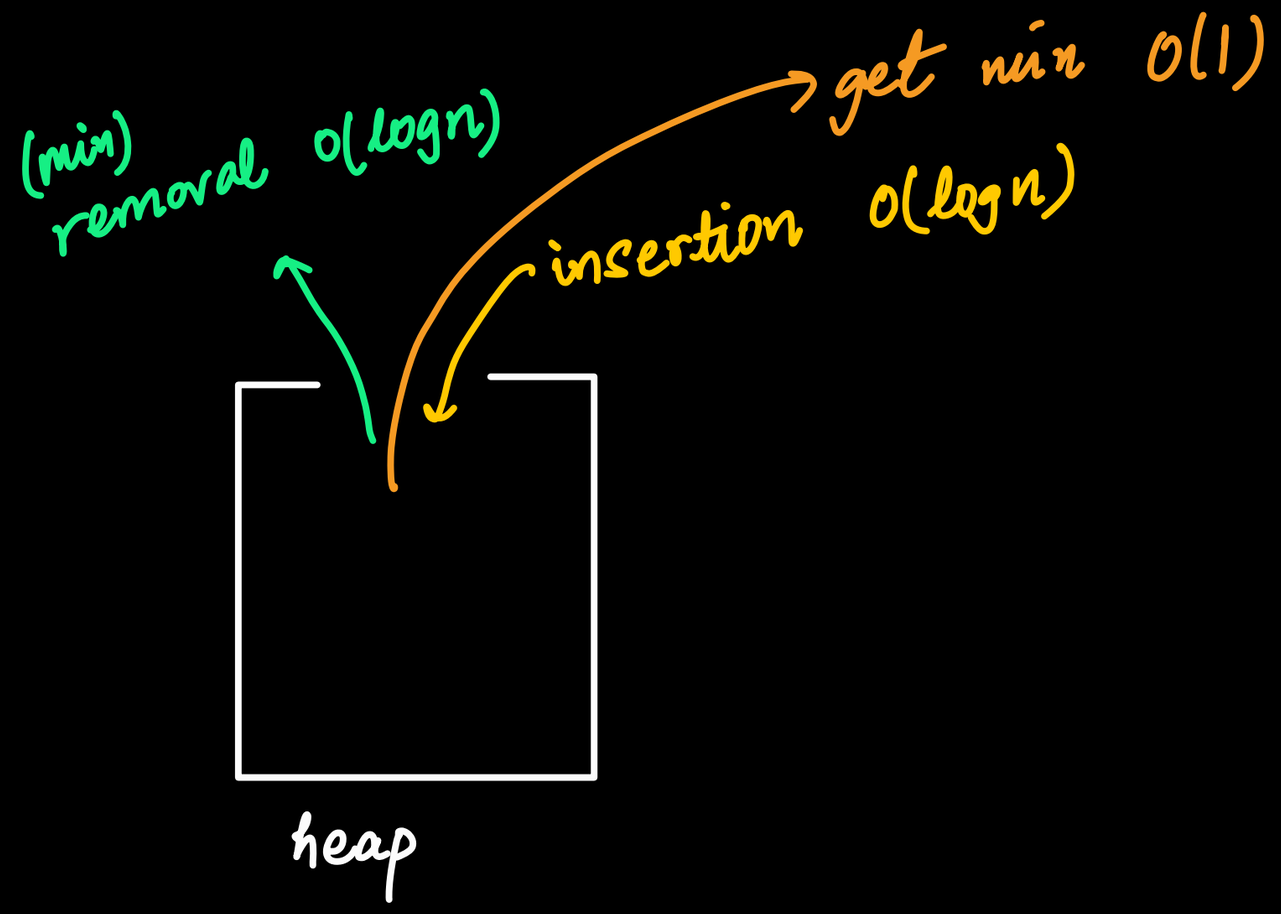
Heaps

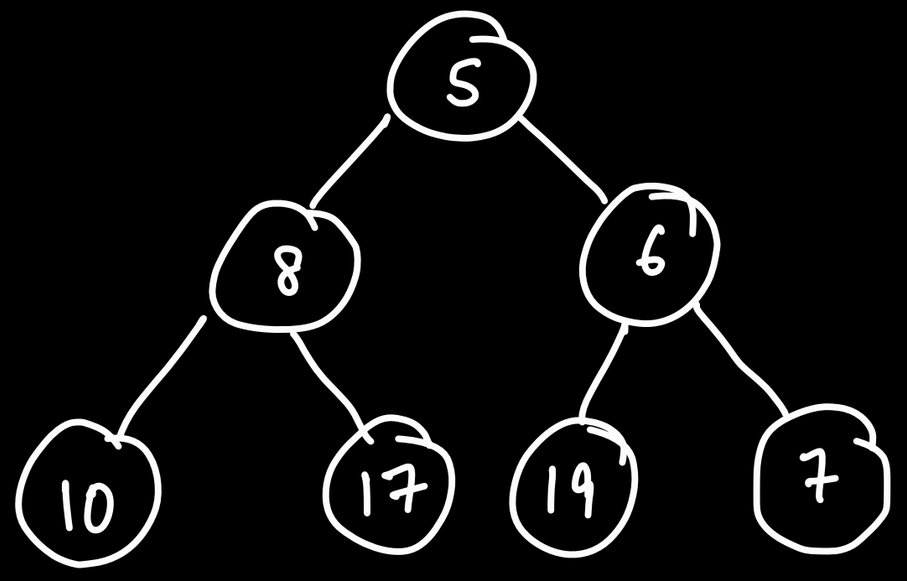
data structure

What is a Heap?



What is a Heap?

Visualised via Binary Tree (CBT)



minheap

What is a Heap?

Kaise pata lage ki sawaal HEAP ka ?

1) k^{th} (smallest, largest, closest, most frequency)

2) Continuous Sorting

Maxheap and Minheap

Minheap:

```
PriorityQueue<Integer> pq = new PriorityQueue<>();
```

Maxheap:

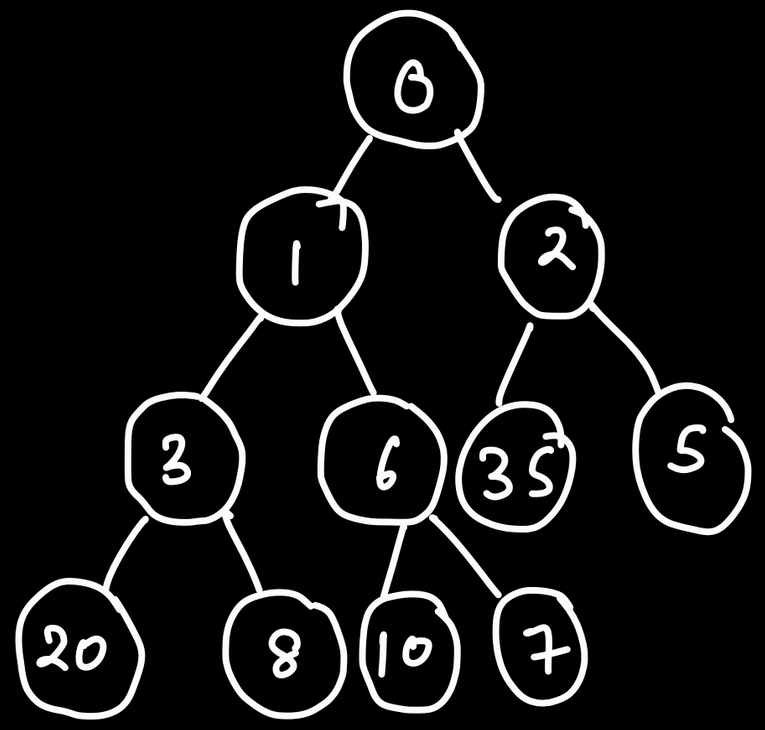
```
PriorityQueue<Integer> pq = new PriorityQueue<>(Collections.reverseOrder());
```

Priority Queue

10 20 -30 35 X

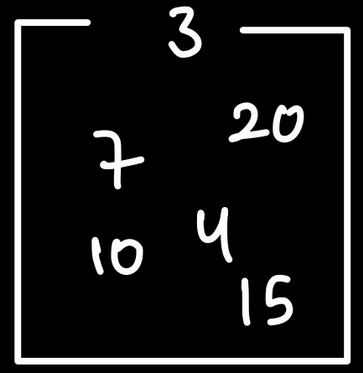
5 8 2 0 3 1 7 6

0 1 2 3 6 35 5 20 8 10 7



Ques: Kth smallest element → max heap

arr = { 7 10 4 3 20 15 } K = 3



minheap

all insertions → n log n

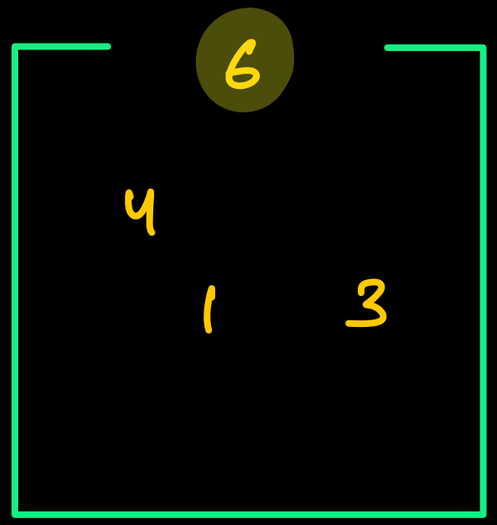
K removals → K log n

total time → (n+K) log n

Ques: Kth smallest element

arr = { 7 10 4 3 20 15 13 1 6 8 } K = 4

13
15
20
8 10
7



maxheap
(size K)

Intuition : We are removing 'n-k' largest elements

insertions → $n \cdot \log K$

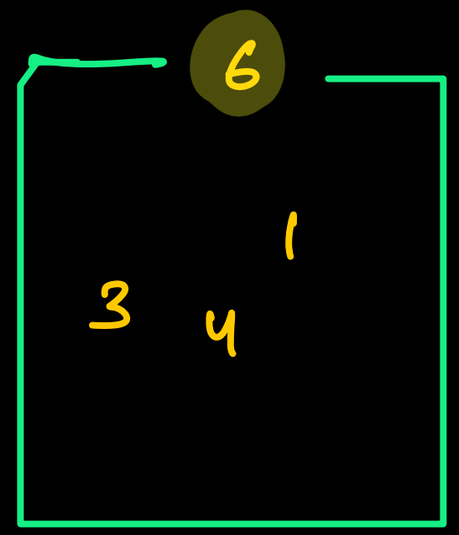
removals → $(n-K) \cdot \log K$

T.C. = $O(n \log K)$

Ques: Kth smallest element

arr = { 1 3 4 6 7 8 10 13 15 20 } K = 4

10
13
8
7
15
20



maxheap
(size ≤ 4)

HW : Kth largest element

minheap

Ques: Sort a K sorted array

arr = { 6 5 3 2 8 10 9 } K = 3

arr = { 2 3 5 6 8 9 10 }

2	3	5	6	8	9	10
↓	↓	↓	↓	↓	↓	↓
+3	+1	-1	-3	0	+1	-1

Ques: Sort a K sorted array

arr = { 6 5 3 2 8 10 9 } K = 3

2 3 5 6 8 9 10

minheap
(size K)

T.C. = O(n log K)

Ques: K closest points to origin

$\{(1,3), (-2,2), (5,4), (0,1)\}$ $K=2$

$\sqrt{10}$ $\sqrt{8}$ $\sqrt{41}$ $\sqrt{1}$

(8, -2, 2)

(1, 0, 1)

maxheap (2)

(41, 5, 4) (10, 1, 3)

HW : Find K closest elements

Ques: Top K frequent elements

HW: Last Stone Weight

Ques: Minimum cost to connect ropes

4 2 7 6 9

6 7 6 9

13 6 9

19 9

28

cost = 6 + 13 + 19 + 28 = 66

4 2 7 6 9

4 2 7 15

4 2 22

4 24

28

cost = 15 + 22 + 24 + 28 = 89

Ques: Minimum cost to connect ropes

2 4 6 7 9

6 6 7 9

12 7 9

19 9

28

cost = 6 + 12 + 19 + 28 = 65

4 2 7 6 9

6 7 6 9

12 7 9

12 16

28

cost = 6 + 12 + 16 + 28 = 62

Ques: Minimum cost to connect ropes

Code : Khud Karo

12

16

28

minheap

cost = 6 + 12 + 16 + 28

Arrays.Sort(arr)

↓

Object ka arr

THANKYOU

Cuties